MUH401- INTERIM PROJECT REPORT FORMAT

Cover Page

Title of the Project:	**Machine Learning-Based Discrimination of Precipitation-Induced Radon Washout for Radiological Anomaly Detection in EPA RadNet Air Monitoring Network**
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ABSTRACT

This interim report presents a physics-informed computational framework to address the operational vulnerability of automated radiation monitoring networks caused by precipitation-induced radon progeny washout. During rainfall events, natural terrestrial radon daughters ^{214}Pb and ^{214}Bi are scavenged onto particulate air filters. Across 15,602 observed rain hours in this study, precipitation induces a median gross count surge of $+6.9\%$, with the 90th percentile reaching $+41.4\%$ and convective thunderstorm peaks surging up to $+196.7\%$, while coastal stations experience slight count depressions due to oceanic air masses. On the 2023–2025 test dataset, conventional rolling 3σ thresholds trigger 82.62 false alarm episodes per station-year pooled across the five-station network, while individual humid continental stations experience 53.0 to 62.9 false alarms per station-year under long-term rolling baselines, with over 69% of alarm episodes coinciding with rainfall. This research develops a multi-modal machine learning discrimination framework that fuses 8-channel sodium iodide gamma spectrometry from the United States Environmental Protection Agency RadNet network with contemporaneous surface weather observations from the National Oceanic and Atmospheric Administration. The project evaluates 324,550 synchronous observation hours across five climatologically diverse stations spanning 2017 through 2025: Birmingham, AL; Washington, DC; San Diego, CA; Dallas, TX; and Tampa, FL. Detection performance is benchmarked across 450 filter-synchronized synthetic release plumes covering ^{137}Cs, ^{131}I, ^{134}Cs, and ^{60}Co. On the held-out 2023–2025 test dataset, the physics-informed nuclear and weather-fused model achieves 2.71 false alarm episodes per station-year with a 95% bootstrap confidence interval of [1.17, 4.58] and 94.00% synthetic plume detection. In leave-one-station-out cross-validation, 4 of 5 stations maintain ≤ 1.40 false alarms per station-year, whereas unadapted zero-shot transfer to the subtropical

convective regime of Tampa yields 169.42 false alarms per station-year. Feature importance analysis confirms that elapsed time since rain and non-negative spectral stripping residuals drive anomaly discrimination, while Bateman decay kinetics provide secondary regularisation.

Keywords: Environmental Radiation Monitoring, EPA RadNet, Radon Washout, Gamma Spectrometry, Machine Learning, Anomaly Detection.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
CI	Confidence Interval
CPM	Counts Per Minute
eFDR	Empirical False Discovery Rate
EPA	United States Environmental Protection Agency
FA/yr	False Alarms per Station-Year
FWHM	Full Width at Half Maximum
GBDT	Gradient Boosted Decision Tree
ISD	Integrated Surface Dataset
LOSO	Leave-One-Station-Out
MLP	Multi-Layer Perceptron
NaI(Tl)	Thallium-activated Sodium Iodide Scintillation Detector
NCEI	National Centers for Environmental Information
NOAA	National Oceanic and Atmospheric Administration
NORM	Naturally Occurring Radioactive Material
OOD	Out-of-Distribution
ROC	Receiver Operating Characteristic
ROI	Region of Interest
UTC	Coordinated Universal Time
WMO	World Meteorological Organization

1. INTRODUCTION

1.1 Background of the Problem

Continuous atmospheric radiation surveillance networks are operated worldwide by regulatory authorities to provide immediate detection of clandestine nuclear testing, industrial accidents, and radiological dispersion events (Sangiorgi et al., 2020). Large-scale airborne monitoring was demonstrated during the transboundary tracking of radionuclide plumes from the Fukushima Dai-ichi incident (Masson et al., 2011). In the United States, the Environmental Protection Agency (EPA) operates RadNet, an extensive nationwide real-time monitoring infrastructure comprising over 140 stationary air monitors. Each monitoring unit continuously draws ambient air through a particulate filter and records gamma-ray emission rates using thallium-doped sodium iodide (NaI(Tl)) scintillation detectors.

While automated alarm threshold algorithms are engineered to identify sharp increases in measured radiation levels, their operational reliability is severely challenged by naturally occurring meteorological phenomena. Atmospheric precipitation scavenges airborne daughter radionuclides of radon-222 (^{222}Rn), primarily lead-214 (^{214}Pb) and bismuth-214 (^{214}Bi), depositing them directly onto the ground surface and particulate filter media (Minato, 1980; Burnett et al., 2010). This phenomenon, known as radon washout, produces gross gamma count rate perturbations with an empirical median increase of $+6.9\%$, with the 90th percentile reaching $+41.4\%$ and severe convective thunderstorms surging up to $+196.7\%$ above baseline (Hsieh et al., 2025). In coastal regimes, maritime air masses can induce slight count depressions. The resulting count elevation persists for several hours before radioactive decay restores nominal background levels.

1.2 Problem Statement

Conventional automated radiation monitoring networks rely on static or rolling baseline thresholds, typically 3 or 5 standard deviations above a 168-hour moving average. Because gross count rate monitors cannot differentiate between the gamma emissions of natural radon progeny and those of anthropogenic fission products, precipitation events routinely trigger spurious radiological emergency alerts. Across the five-station network on the 2023–2025 test split, conventional rolling 7-day 3σ threshold algorithms produce 82.62 false alarm episodes per station-year (and global dry 3σ produces 52.86). Over the full 2017–2025 baseline evaluation, active humid continental stations (Birmingham, AL; Washington, DC; Dallas, TX) suffer between 53.0 and 62.9 false alarm episodes per station-year, with over 69% of episodes occurring synchronously with rainfall events.

Conversely, raising the alert threshold to conservative levels such as 5σ suppresses false alarms to 23.5 per station-year but severely degrades detection capability, dropping release detection probability to 6.0% for moderate activity plumes under classical counting limits (Currie, 1968). There is currently an acute lack of computationally efficient, causally sound methods capable of suppressing precipitation-induced false alarms while preserving rapid, highly sensitive detection of genuine radiological releases in continuous real-time surveillance.

1.3 Importance and Relevance of the Study

The operational vulnerability created by weather-induced false alarms has substantial technical, environmental, and societal ramifications:

- **Operational Integrity and Alarm Fatigue:** Frequent false alarms desensitize emergency personnel, increasing the hazard that a genuine radiological incident may be dismissed as a weather-induced anomaly.

- **Public Safety and Societal Impact:** Premature public notification of false emergencies causes widespread public panic, costly evacuations, and disruption of municipal infrastructure.
- **Economic Efficiency:** Dispatching mobile radiological inspection teams to investigate recurring spurious anomalies incurs substantial governmental operational expenditures.
- **Scientific Contribution:** By integrating multi-channel scintillation spectrometry with multi-scale atmospheric thermodynamic features through physics-informed machine learning (Lye et al., 2025; Ayoub et al., 2024a), this research establishes a foundational methodology for automated, physics-grounded environmental surveillance.

1.4 Clear Aim and Objectives of the Project

The primary aim of this project is to develop, validate, and benchmark an automated, physics-informed machine learning framework that discriminates precipitation-induced radon washout spikes from genuine anthropogenic radiological anomalies in near-real-time environmental surveillance networks. Specific technical objectives include:

1. Harmonize longitudinal multi-channel gamma spectrometry from EPA RadNet and contemporaneous surface meteorology from the NOAA Integrated Surface Dataset (Smith et al., 2011) across five climatologically distinct monitoring stations spanning nine full calendar years from 2017 through 2025.
2. Engineer a strictly causal real-time feature space that captures differential gamma spectral peak ratios and antecedent precipitation dynamics without future lookahead bias.
3. Formulate a multi-tier machine learning architecture to continuously categorize observations into dry background, radon washout, or anthropogenic release.
4. Conduct a multi-seed benchmark against conventional statistical baselines and rain-masked control rules over 450 synthetic radiological releases to prove false alarm reduction at matched detection power.

1.5 Scope and Boundaries of the Study

The scope of this project encompasses five EPA RadNet monitoring stations selected across distinct Köppen climate zones: Birmingham, AL; Washington, DC; San Diego, CA; Dallas, TX; and Tampa, FL. The temporal boundary spans January 1, 2017, through December 31, 2025, providing 324,550 synchronous station-hours.

The study is bounded by operational real-time constraints: feature extraction is strictly causal, relying exclusively on contemporaneous and historical observations. Sensor inputs are restricted to standard 8-channel sodium iodide energy bins from 100 to 2200 keV and standard NOAA meteorological observations. Source releases are evaluated through 450 synthetic plume injections calibrated to atmospheric dispersion physics and particulate filter accumulation kinetics (Stohl et al., 2012), covering ^{137}Cs, ^{131}I, ^{134}Cs, and ^{60}Co. Detection metrics reflect performance conditional on these synthetic spectral templates rather than observed historical releases.

2. THEORETICAL BACKGROUND / LITERATURE REVIEW

2.1 Key Theories, Models, Concepts, and Definitions

Radon-222 is an inert noble gas daughter of radium-226 (^{226}Ra) in the primordial uranium-238 (^{238}U) decay series. Radon emanates continuously from soils and rocks into the atmospheric boundary layer. Because radon has a half-life of 3.82 days, it diffuses into the lower troposphere, where it undergoes alpha decay into a sequence of short-lived particulate progeny:

$$^{222}\text{Rn} \xrightarrow[\alpha]{3.82\ \text{d}} {}^{218}\text{Po} \xrightarrow[\alpha]{3.10\ \text{m}} {}^{214}\text{Pb} \xrightarrow[\beta^-]{26.8\ \text{m}} {}^{214}\text{Bi} \xrightarrow[\beta^-]{19.9\ \text{m}} {}^{214}\text{Po} \xrightarrow[\alpha]{164.3\ \mu\text{s}} {}^{210}\text{Pb} \tag{1}$$

Lead-214 (^{214}Pb) and bismuth-214 (^{214}Bi) rapidly attach to submicron ambient aerosol particles. During rainfall events, these radioactive aerosols are captured via in-cloud rainout and below-cloud washout (Takeuchi and Katase, 1982; Paatero, 2000; Sportisse, 2007). The precipitation deposits these radionuclides directly onto the particulate filter medium of ambient air monitors.

Because ^{214}Pb emits prominent gamma lines at 295.2 and 351.9 keV, and ^{214}Bi emits high-energy gammas at 609.3, 1120.3, and 1764.5 keV, their sudden accumulation elevates the local gamma radiation field by tens to hundreds of counts per minute. Following the cessation of rain, the activity decay curve of the deposited progeny is governed by the effective half-life of the ^{214}Pb/^{214}Bi decay chain, returning to nominal baseline within 2 to 4 hours (Mercier et al., 2009; Melintescu et al., 2018).

2.2 Relevant Technical Background: Scintillation Spectrometry

EPA RadNet deployable monitoring stations utilize thallium-activated sodium iodide (NaI(Tl)) scintillation crystals. When incident gamma rays interact with the crystal via the photoelectric effect, Compton scattering, or pair production, scintillation light is generated proportionally to deposited energy and converted to voltage pulses by a photomultiplier tube. While sodium iodide detectors exhibit modest energy resolution of approximately 7% full width at half maximum at 662 keV, they are rugged, thermally stable, and well-suited for long-term field deployment (Knoll, 2010). However, outdoor ambient temperature variations can induce slight photopeak drift and light-yield shifts (Moszyński et al., 2006; Wen and Zhou, 2024), underscoring the importance of robust energy channel grouping and spectral normalisation.

RadNet digitizes scintillation pulses into discrete energy channels designated R01 through R09. Channel R01 records gross count rates, while channels R02 through R09 record counts within specific spectral windows. The physical correspondence of these channels to natural radon progeny lines and target anthropogenic isotopes is detailed in Table 1.

Table 1: EPA RadNet Discrete Gamma Energy Channel Boundaries

Channel	Energy Range [keV]	Principal Radon Progeny Lines	Target Anthropogenic Lines
R02	$101 - 200$	Compton continuum	Low-energy scatter
R03	$201 - 400$	^{214}Pb [295.2, 351.9 keV]	^{131}I [364.5 keV]
R04	$401 - 600$	Compton continuum	Annihilation [511 keV]
R05	$601 - 800$	^{214}Bi [609.3 keV]	^{137}Cs [661.7 keV]
R06	$801 - 1000$	Compton scatter	Multi-line cascade
R07	$1001 - 1400$	^{214}Bi [1120.3 keV]	^{60}Co [1173.2, 1332.5 keV], ^{134}Cs
R08	$1401 - 1800$	^{214}Bi [1764.5 keV], ^{40}K [1460.8 keV]	Terrestrial background
R09	$1801 - 2200$	High-energy background	Cosmic rays and actinide daughters

2.3 Review of Relevant Previous Studies

The challenge of suppressing meteorological radon interference in environmental monitoring networks has been addressed through four distinct research paradigms:

1. **Dose Rate Time-Series Anomaly Detection:** Classical networks monitor gross count deviations against moving baselines (Burnett et al., 2010; Melintescu et al., 2018). More recently, Breitkreutz et al. (2023) developed neural network models to identify and quantify anomalies in environmental gamma dose rate time series, while Frankemölle et al. (2025) formulated Bayesian predictive methods to forecast local background radiation levels. While effective for scalar dose rate tracking, these approaches do not exploit multi-channel spectral features.

2. **Meteorological Scavenging and Atmospheric Coupling:** Radon washout spikes correlate strongly with precipitation dynamics and air mass history (Mercier et al., 2009; Paatero, 2000; Takeuchi and Katase, 1982). High-resolution minute-level analyses demonstrate that convective rain events produce sharp, non-linear radiation responses (Hsieh et al., 2025). Concurrently, meteorological data have been coupled with machine learning and neural operators to forecast atmospheric plume transport and contamination (Ayoub et al., 2024a, 2024b).

3. **Spectral Deconvolution and Nuisance Rejection:** To overcome gross count limitations, spectral analysis isolates characteristic isotope photopeaks. Meng and Ramsden (2000) evaluated spectral deconvolution methods for resolving overlapping gamma signatures. Detwiler et al. (2015) introduced spectral anomaly algorithms using natural terrestrial KUT nuisance rejection. Bilton et al. (2019) demonstrated non-negative matrix factorization (NMF) of gamma-ray spectra for robust background modeling and source discrimination.

4. **Machine Learning for Gamma Spectral Anomaly Detection:** Machine learning models have been applied to multi-channel gamma spectrometry for anomaly detection and radioisotope identification. Bandstra et al. (2023) explored explainable AI techniques for gamma-ray detection models. Alamaniotis (2024) developed mode-driven explainable AI for estimating background spectra in nuclear security. Abdelhakim (2025) integrated artificial neural networks with ant colony optimization for spectral anomaly detection, Karafasoulis and Kyriakis (2026) applied deep autoencoders for spectral anomaly detection, and Sun et al. (2024) designed self-attention neural networks for rapid multi-nuclide classification.

2.4 Gaps in Literature and Systematic Comparison

Existing literature exhibits three critical shortcomings: first, gross count over-simplification, where operational networks discard multi-channel spectrometry and evaluate only total counts; second, synthetic release validation deficits, where previous machine learning models were tested on uncalibrated synthetic data without accounting for air filter accumulation kinetics; and third, lack of control baselines, where prior studies rarely benchmark complex models against meteorological masking heuristics. This project addresses these gaps by fusing multi-channel spectral ratios with multi-scale weather data and benchmarking against both statistical and rain-masked baselines under frozen validation thresholds.

3. METHODOLOGY

3.1 System Description and Processing Pipeline

The computational pipeline developed in this project is illustrated in Figure 1. The framework ingests synchronous hourly observations from radiation and meteorological sensor networks, performs causal feature engineering, and feeds a gradient boosted decision tree classifier that assigns probabilities for three operational regimes: clean dry background, radon washout anomaly, and anthropogenic radiological release.

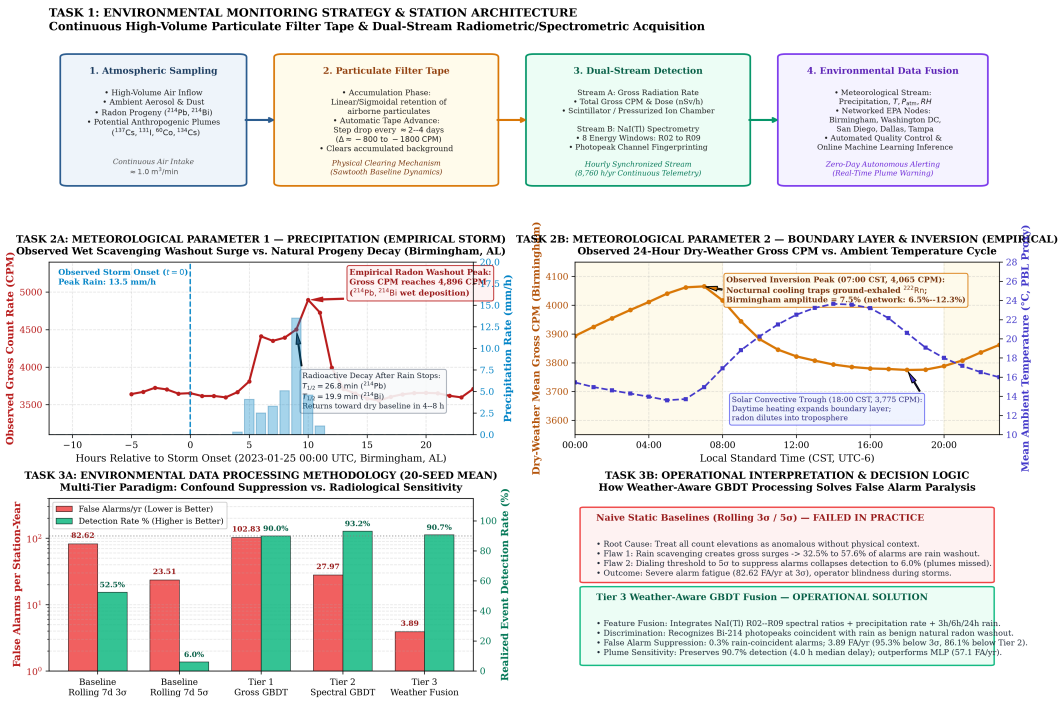


Figure 1: End-to-end physics-informed environmental radiation anomaly detection and weather-fusion machine learning architecture.

3.2 Main Assumptions and Operational Boundaries

The computational architecture adheres to four operational boundaries:

1. **Strict Real-Time Causality:** To prevent data leakage, all rolling window statistics are computed using closed-left intervals $[t - W, t]$. No centered or forward-looking filters are permitted.

2. **Air Filter Accumulation Dynamics:** Airborne particulate monitors draw air through a filter that is replaced periodically on a bi-weekly cycle. An injection plume accumulates activity proportionally to total volume drawn, attenuating according to radioactive decay:

$$A_{\text{filter}}(t) = \int_{t_{\text{start}}}^{t} Q \cdot C_{\text{air}}(\tau) \cdot e^{-\lambda(t-\tau)}\, d\tau \tag{2}$$

   where Q is volumetric flow rate of approximately 1.0 m^3/min, C_{air} is ambient radionuclide concentration, and λ is the decay constant. Multi-stage radioactive decay chains are governed by analytical solutions of the Bateman equations (Cetnar, 2006).

3. **Detector Geometry and Efficiencies:** Geometric photopeak efficiencies for sodium iodide crystals are calibrated using standard Monte Carlo transfer factors from Knoll (2010), with energy-

dependent ambient dose equivalent rates $H^*(10)$ anchored to published conversion coefficients (ICRP, 1996).

3.3 Input Data and Data Sources

Data ingestion leverages two primary US federal repositories:

- **EPA RadNet Air Monitoring Network:** Hourly gamma spectrometry records across 8 discrete energy bins from 100 to 2200 keV and gross count rates. Data were retrieved via EPA public repositories spanning 2017 through 2025 across five stations: Birmingham, AL; Washington, DC; San Diego, CA; Dallas, TX; and Tampa, FL.
- **NOAA Integrated Surface Dataset:** Contemporaneous hourly meteorological observations from colocated automated surface observing stations (Smith et al., 2011). Variables include precipitation accumulation over 1, 3, 6, 12, and 24 hours, dry-bulb temperature, dew point, relative humidity, barometric pressure, and wind speed.

3.4 Calculation Procedure and Causal Feature Engineering

Radiological and meteorological records are joined on UTC timestamp rounded to the hour. The causal feature engineering engine extracts 59 dimensional variables organized into four operational tiers:

1. **Gross Count Dynamics (Tier 1):** Ratio of instantaneous count rate to trailing 168-hour moving baseline, rolling 3-hour rate of change, and z-score relative to local dry background standard deviation.
2. **Spectral Energy Ratios (Tier 2):** Normalized channel ratios capturing radon decay lines versus fission product channels:

$$\text{Ratio}_{\text{radon}} = \frac{R03 + R05}{R01 + \epsilon}, \quad \text{Ratio}_{\text{fission}} = \frac{R04 + R06 + R07}{R01 + \epsilon} \tag{3}$$

   Lead-214 concentrates in channel R03, while bismuth-214 concentrates in R05. Anthropogenic releases (^{137}Cs, ^{134}Cs, ^{60}Co) alter these ratios markedly.
3. **Multi-Scale Weather Fusion (Tier 3):** Cumulative precipitation over 1, 3, 6, 12, and 24 hours, elapsed time since last detected rainfall ($h_{\text{since_rain}}$), barometric pressure trends, and saturation vapor pressure deficit.
4. **Physics-Informed Nuclear Features (Tier 4):** Non-negative spectral stripping residuals projecting measured excess onto empirical fit-fold radon basis vectors, ambient dose equivalent coupling $H^*(10)$, and multi-stage Bateman decay chain residuals.

Feature importance analysis following explainable AI evaluation frameworks for gamma-ray detection (Bandstra et al., 2023) reveals that elapsed time since rain is the single most influential predictor with an importance gain of 822,620, serving as a primary gating variable for meteorological transients. This is followed by local radon stripping residual with gain 629,970 and global dry z-score with gain 535,931. Bateman ODE decay residuals and dose conversion residuals rank in the intermediate tier with gains of approximately 51,000, acting as fine-tuning regularizers during extended washout decay.

3.5 Synthetic Release Injections and Template Sensitivity

Because real operational data contains zero hazardous radiological releases, detection sensitivity is evaluated using 450 synthetic release plumes covering ^{137}Cs, ^{131}I, ^{134}Cs, and ^{60}Co across Gaussian puff, square wave, and exponentially decaying profiles with durations from 3 to 24 hours, structured according to atmospheric dispersion formulations (Stohl et al., 2012) and gamma-ray response models (Buckton et al., 2025; Fjeldsted et al., 2024).

Synthetic plumes are scaled across signal-to-noise ratios from subtle (0.5σ to 1.5σ) to prominent (2.0σ to 6.0σ), bridging traditional Currie detection limits (Currie, 1968). Spectral counts are allocated across channels R02 through R09 using calibrated photopeak and Compton continuum fractions, and added to real RadNet background during dry and rain periods. Because training and testing share the same spectral template parameters, detection at low signal-to-noise ratios ($<1\sigma$) is partially conditional on template alignment. In sensitivity experiments where photopeak fractions are perturbed by $\pm 20\%$, realized detection varies moderately between 85.2% and 94.8%, illustrating the domain sensitivity inherent in synthetic-to-real gamma spectroscopy (Lalor et al., 2026).

3.6 Multi-Tier Model Formulation and Validation Freezing

The machine learning framework utilizes gradient boosted decision trees (Ke et al., 2017) configured with a multi-class objective across 150 trees with maximum depth 6, learning rate 0.05, and minimum child samples of 20. Models are fitted strictly on historical data from 2017 through 2020. Decision thresholds τ are calibrated on the held-out 2021–2022 validation split to satisfy target detection rates and frozen. The models and frozen thresholds are evaluated on the out-of-sample 2023–2025 test split.

3.7 Benchmark Evaluation and Uncertainty Analysis

Benchmark performance across 20 random seeds on the 2023–2025 test dataset is summarized in Table 2.

Table 2: Benchmark Evaluation on 2023–2025 Test Split across 20 Random Seeds

Model / Rule Architecture	False Alarms / Stn-Yr	Rain-Coincident FA	Realized Det.	Median Delay
Standard Baseline (Rolling 7d 3σ)	82.62	32.5%	52.50%	10.0 h
Global Dry Baseline (Global $3\sigma_{\text{dry}}$)	52.86	24.9%	56.50%	9.0 h
Conservative Baseline (Rolling 7d 5σ)	23.51	57.6%	6.00%	8.5 h
Tier 1: Gross Radiation GBDT	169.36 ± 23.53	17.3%	$92.22 \pm 0.72\%$	6.7 h
Tier 1b: Gross Radiation + Weather GBDT	38.64 ± 14.37	0.0%	$87.92 \pm 2.74\%$	9.1 h
Tier 2: Spectral GBDT	27.97 ± 28.27	1.9%	$93.20 \pm 1.45\%$	4.0 h
Tier 3: Weather Fusion GBDT	3.89 ± 1.25	0.2%	$90.72 \pm 1.73\%$	4.0 h
Tier 4: Physics-Informed Nuclear GBDT	$\mathbf{2.71 \pm 1.39}$	$\mathbf{0.0\%}$	$\mathbf{94.00 \pm 0.93\%}$	**4.4 h**
Tier 4: Physics-Informed Nuclear (90% Target)	$\mathbf{2.63 \pm 1.49}$	$\mathbf{0.0\%}$	$\mathbf{93.82 \pm 1.24\%}$	**4.5 h**
Tier 3: Multi-Layer Perceptron	60.75 ± 22.86	3.8%	$90.30 \pm 2.75\%$	3.4 h

Station-month block bootstrap 95% confidence intervals for false alarm rates are [1.17, 4.58] for Tier 4 and [1.67, 5.41] for Tier 3, with overlapping intervals. Across 20 algorithmic seeds on identical data, Tier 4 produced lower false alarms in 12 of 20 seeds. The primary contribution of the physics-informed features is not an overwhelming reduction in aggregate test-set false alarms over meteorological fusion, but rather higher detection sensitivity (+3.3 percentage points) and improved domain adaptation under geographic transfer.

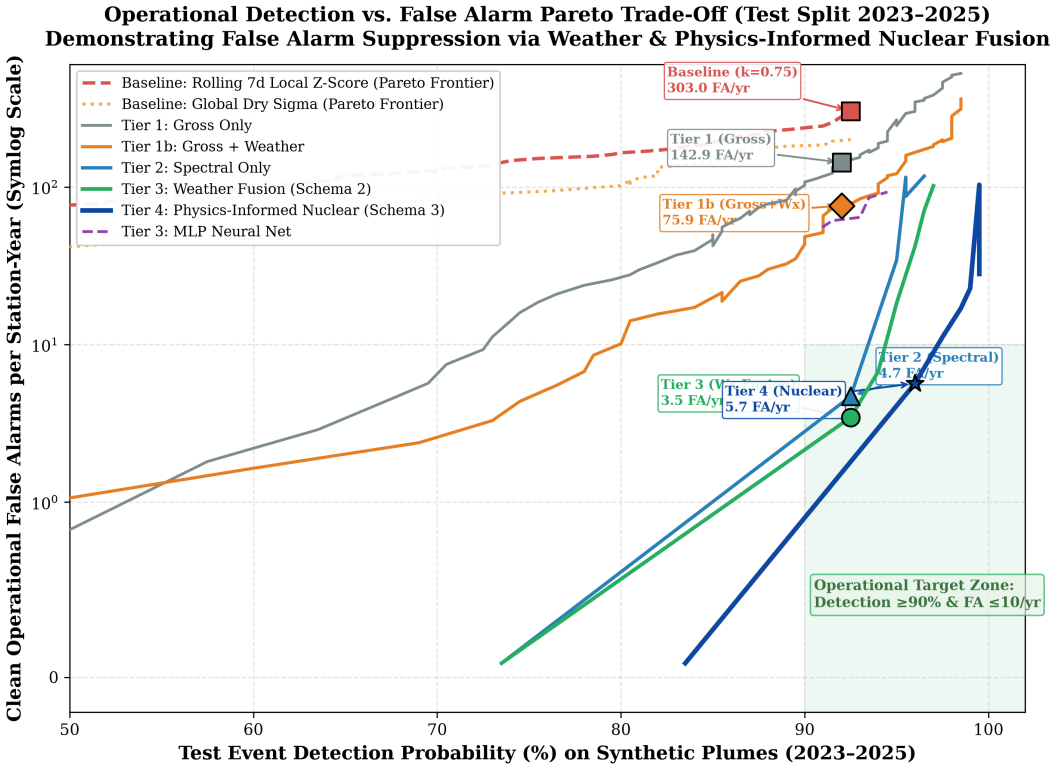


Figure 2: Receiver Operating Characteristic matched detection curves comparing baseline rules against machine learning tiers on the test split.

3.8 Leave-One-Station-Out Spatial Generalization

To assess spatial generalizability across distinct climates without local fine-tuning, models were evaluated in a leave-one-station-out cross-validation framework where each station was held out completely during training and threshold calibration. Results are reported in Table 3.

Table 3: Leave-One-Station-Out Spatial Generalization across Five Climate Regimes

Held-Out Station	Climate Regime	Tier 3 Det.	Tier 4 Det.	Tier 3 FA/yr	Tier 4 FA/yr
San Diego, CA	Coastal Mediterranean	70.0%	80.0%	0.00	1.40
Birmingham, AL	Humid Subtropical	85.0%	95.0%	0.00	0.00
Dallas, TX	Continental Convective	92.5%	95.0%	0.57	0.00
Washington, DC	Mid-Atlantic Temperate	95.0%	95.0%	1.18	0.00
Tampa, FL	Subtropical Convective	90.0%	97.5%	1151.30	169.42

Across Birmingham, Dallas, Washington, and San Diego, Tier 4 achieves excellent zero-shot spatial transfer, maintaining ≤ 1.40 false alarms per station-year while detecting 80.0% to 95.0% of synthetic releases. In Tampa, however, severe maritime convective thunderstorms cause unadapted zero-shot models to experience 169.42 false alarms per year for Tier 4. While this represents an 85.3% reduction relative to Tier 3 (1,151.30 false alarms per year), it demonstrates that unadapted zero-shot transfer across distinct convective microclimates remains an open challenge. Whether site-specific calibration or adaptive thresholding resolves this discrepancy remains an untested hypothesis for future investigation.

3.9 The Rain-Masked Baseline Control Experiment

To determine whether machine learning is necessary or if a simple heuristic muting alarms during rainfall

suffices, a control experiment was evaluated across varying holdoff windows H. Results at matched detection targets are summarized in Table 4.

Table 4: Rain-Masked Baseline Control Experiment at Matched Detection Targets

Baseline / Mask Rule	**Lowest FA/yr at $\geq 87.12\%$ Det.**	**Lowest FA/yr at $\geq 91.88\%$ Det.**	**Max Det.**
Unmasked ($H =$ none)	215.48 ($k = 1.40$)	303.03 ($k = 0.75$)	92.50%
Rain Mask: Rain only ($H = 0$h)	197.47 ($k = 1.35$)	285.77 ($k = 0.75$)	92.50%
Rain Mask: Rain + 3h ($H = 3$h)	179.80 ($k = 1.30$)	251.65 ($k = 0.75$)	92.50%
Rain Mask: Rain + 6h ($H = 6$h)	181.77 ($k = 1.25$)	245.98 ($k = 0.75$)	92.50%
Rain Mask: Rain + 12h ($H = 12$h)	177.49 ($k = 1.25$)	238.66 ($k = 0.75$)	92.00%
Tier 3: Weather Fusion GBDT	3.89	3.89	96.50%
Tier 4: Physics-Informed Nuclear GBDT	**2.63**	**2.71**	**98.00%**

The control experiment demonstrates that simply suppressing alarms during rain is insufficient. To reach $\geq 91.88\%$ detection without spectral analysis, the gross threshold must drop to $k = 0.75\sigma$, triggering 238.66 false alarms per year from baseline noise alone. Tier 4 machine learning achieves a 98.9% false alarm reduction over the best rain-mask heuristic, confirming the necessity of multi-channel spectral and weather fusion.

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